A STANDARD COVERING AND THE SINGLE MORPHISM FROM THE DISJOINT UNION GIVE ISOMORPHIC CECH COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement has three cases according to the site considered – the big site for one of five topologies, the small etale site, the small Zariski site – and the proof treats them uniformly; the case distinction is reproduced as delivered.

Editorial note. The statement refers to the map of Cech complexes defined in the cohomology-on-sites chapter by an equation label; that reference lies outside the chapter and is carried as an external reference stub, so its number resolves without the equation being reproduced.

Editorial note. The counted spans carry 1 navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Standard affine coverings in a topology and the topologies considered. Topologies, the definition of a standard affine covering, used in the statement at lines 2915-2916. Fixes the object whose Cech complex is compared with the one of the single covering morphism.
- The map of Cech complexes induced by a refinement. Cohomology on Sites, Equation sites-cohomology-equation-map-cech-complexes, invoked in the statement at lines 2921-2923. Fixes what the isomorphism asserted in the statement is an isomorphism of.
- Fibre products commute with disjoint unions, and a sheaf takes disjoint unions to products. Standard, used at proof lines 2934-2944. Computes both Cech complexes and shows the comparison map is an isomorphism.

Source reference. Original equation 03F4: sources/stacks-chow/tags.txt, source lines 4479–4479. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let*

(a) *$\tau \in \{Zariski, fppf, étale, smooth, syntomic\}$ and $\mathcal{C} = (Sch/S)_\tau$, or*
(b) *let $\tau = étale$ and $\mathcal{C} = S_{étale}$, or*

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(c) *let $\tau = Zariski$ and $\mathcal{C} = S_{Zar}$.*

Let $\mathcal{F}$ be an abelian sheaf on $\mathcal{C}$. Let $U \in \mathrm{Ob}(\mathcal{C})$ be affine. Let $\mathcal{U} = \{U_i \to U\}_{i=1,\ldots,n}$ be a standard affine τ-covering in $\mathcal{C}$. Then

(1) *$\mathcal{V} = \{\coprod_{i=1,\ldots,n} U_i \to U\}$ is a τ-covering of U,*
(2) *$\mathcal{U}$ is a refinement of $\mathcal{V}$, and*
(3) *the induced map on Čech complexes (Cohomology on Sites, Equation (03F4))*

$$\check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{F}) \longrightarrow \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$$

is an isomorphism of complexes.

Proof. This follows because

$$(\coprod\nolimits_{i_0=1,\ldots,n} U_{i_0}) \times_U \ldots \times_U (\coprod\nolimits_{i_p=1,\ldots,n} U_{i_p}) = \coprod\nolimits_{i_0,\ldots,i_p \in \{1,\ldots,n\}} U_{i_0} \times_U \ldots \times_U U_{i_p}$$

and the fact that $\mathcal{F}(\coprod_a V_a) = \prod_a \mathcal{F}(V_a)$ since disjoint unions are τ-coverings. □